Joint Distribution Adaptive-Alignment for Cross-Domain Segmentation of High-Resolution Remote Sensing Images

Haitao Huang, Baopu Li, Yuchen Zhang, Tao Chen, *Senior Member, IEEE*, and Bin Wang, *Senior Member, IEEE*

***Abstract*— Although existing unsupervised domain adaptation (UDA) methods have successfully applied to semantic segmentation tasks for high-resolution remote sensing (HRS) images, they still have some limitations that need to be addressed: 1) they mainly focus on aligning the marginal distributions while ignoring the interdomain differences in the conditional distributions, which may be suboptimal because they assume that the boundaries of category decision are identical across domains; and 2) they depend on self-supervised learning for easy-to-hard alignment, which may result in model learning erroneous knowledge from the pseudo labels. To address the above limitations, we propose a joint distribution adaptive-alignment framework (JDAF) to eliminate the distribution difference between the source and target domains, which is mainly composed of a marginal distribution alignment (MDA) module, a conditional distribution alignment (CDA) module, and an improved easy-to-hard adaptation strategy. The MDA module is used to narrow local semantic and global spatial differences between domains and first advance, and then, the CDA module that includes a category-invariant feature alignment (CFA) block and a dataset-level context aggregation (DCA) block is presented and designed, which can dynamically update and align the feature representations that are invariant to category change and adaptively incorporate dataset-level context into the features of source domain to enhance the pixel-level representation. An uncertainty-adaptive learning (UAL) method is, moreover, proposed to improve the easy-to-hard adaptation strategy by enabling the model to learn accurate knowledge from the pseudo labels, which can boost the adaptive performance of the whole JDAF. Comprehensive experiments with four cross-domain tasks on two benchmark datasets of aerospace HRS images demonstrate that the proposed JDAF achieves significant performance gains compared to the state-of-the-art cross-domain semantic segmentation methods. Our code is available at: https://github.com/maple-hx/JDAF.**

***Index Terms*— Dataset-level context aggregation (DCA), high-resolution remote sensing (HRS) images, joint distribution alignment, semantic segmentation, uncertainty-adaptive learning (UAL), unsupervised domain adaptation (UDA).**

I. Introduction

HIGH-RESOLUTION remote sensing (HRS) images provide rich and detailed information about the Earth's surface, making them valuable for many applications, such as traffic management, land surveying, urban planning, and environmental monitoring [1], [2], [3]. Semantic segmentation, which assigns a land cover category (such as building, tree, etc.) to each pixel in an image, plays a critical role in the effective analysis of HRS images.

Traditional methods for the segmentation of HRS images rely on hand-crafted features and the expertise of domain experts [4], [5], and they may not capture detailed structural information with the increasing spatial resolution of images. In recent years, CNN-based approaches have shown remarkable performance in semantic segmentation [6], [7], [8], [9], [10], [11], [12], [13], [14], but they require large amounts of labeled data for training. Collecting and annotating pixel-level labeled datasets is usually time-consuming and expensive. Hence, it is essential to develop a general segmentation model that can accurately predict category probabilities across multiple datasets without requiring large amounts of labeled data; however, the performance of these models is significantly affected by the distribution differences between the training and test datasets. In the actual situation, the imaging conditions of different HRS datasets, such as geographic location, imaging sensors, and imaging time seasons, often result in significant changes in the data distributions, making it challenging to transfer the segmentation models from a well-labeled source domain to an unlabeled target domain. Thus, domain adaptation methods are needed to transfer the segmentation networks between different HRS datasets.

To achieve the above goal, unsupervised domain adaptation (UDA) can be used to reduce domain differences and learn invariant representations between labeled source domains and unlabeled target domains [15]. Metric discrepancy-based methods minimize feature distribution differences between domains [16], [17]. Adversarial adaptive approaches generate adversarial losses at different levels (including pixel, feature, and output levels [20], [21], [22], [23], [24], [25], [26], [27], [28]) to reduce domain discrepancies in terms of image styles, deep feature space, and category probability prediction output space. Despite the impressive performance of these domain-adaptive methods in cross-domain semantic segmentation, they are tailored for natural image datasets. These methods often fail to achieve satisfactory results when applied to the semantic segmentation of HRS images, mainly because HRS images have significant differences from natural images in various aspects such as shooting angle, spatial complexity, and image resolution, resulting in greater intraclass differences

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and interclass similarities in features extracted from HRS images. In addition, HRS images are often very large in size, so they need to be cropped or downsampled, which may result in loss of information or discontinuous boundaries. These differences between HRS images and natural images can cause additional difficulties in processing HRS images.

Although UDA has also been applied in the semantic segmentation of HRS images [29], [30], [31], [32], [33], current UDA methods only align marginal distributions between domains and treat the feature distributions of the source and target domains as unlabeled clustering structures for adaptation, which is insufficient to obtain optimal adaptation for the segmentation network when distribution differences between domains are significant. Existing methods based on contextual information aggregation [34], [35], [36], moreover, obtain contextual information from one batch of images, which cannot represent that of the whole dataset. Besides, UDA works based on self-supervised and adversarial learning (such as IntraDA [28] and CCDA [33]) apply a two-stage adaptation method to the target domain, which is divided into easy and hard subsets based on the adaptation difficulty. The first stage uses the easy subset, while the second stage applies the hard subset; however, this method may introduce false knowledge into the model, resulting in damage to model adaptation.

To solve the above problems, we propose a joint distribution adaptive-alignment framework (JDAF) in this article for semantic segmentation of HRS images. It mainly includes a marginal distribution adaptation (MDA) module, a conditional distribution adaptation (CDA) module, and an improved easy-to-hard adaptation strategy for both MDA and CDA. The MDA module is used to reduce cross-domain feature discrepancies from local details and global spatial information to align interdomain marginal distributions. The CDA module, composed of a category-invariant feature alignment (CFA) block and a dataset-level context aggregation (DCA) block, is designed and built to address the inconsistent category decision boundaries and underused contextual information. It extracts and aligns category-invariant features and fuses dataset-level context with source features to enhance category representation; moreover, an improved strategy of two-stage easy-to-hard adaptation is constructed. Specifically, to prevent the model from learning erroneous knowledge in the pseudo labels in the second stage of the easy-to-hard adaptation, an uncertainty-adaptive learning (UAL) method is suggested, consisting of uncertainty-adaptive self-supervised learning and uncertainty-adaptive adversarial learning. The uncertainty-adaptive self-supervised learning is used to realize noise filtering by means of the pixel-level weighting of the pseudo labels, and the uncertainty-adaptive adversarial learning is used to further adapt the low-confidence samples among the easy-to-adapt samples by using the sum of the pixel entropy values to measure the difficulty of each target image. Extensive experimental results show that the proposed JDAF achieves new state-of-the-art performance in cross-domain semantic segmentation for HRS images.

The main contributions of this article can be concisely summarized as follows.

1) The proposed JDAF is the first attempt to explicitly apply both MDA and CDA in UDA for semantic segmentation of HRS images. Specifically, the MDA and CDA modules are integrated to achieve the above goal, greatly boosting the performance of cross-domain alignment.
2) In the proposed CDA module, the CFA block is designed to align interdomain category decision boundaries and the DCA block is constructed to adaptively enhance feature representation through category attention driving so as to implement the alignment of conditional distributions and accurate semantic segmentation.
3) To address the issue of false knowledge in pseudo labels, the UAL method, including uncertainty-adaptive self-supervised learning and adversarial learning, is advanced to enable our architecture to learn better domain representations.

The remainder of this article is organized as follows. Section II reviews the works on semantic segmentation, UDA for semantic segmentation, and UDA for semantic segmentation in HRS images. Section III presents the JDAF (including the MDA and CDA modules and the UAL method) and describes its network structure, optimization objectives, and joint adaptation strategy in detail. Section IV evaluates the performance of the proposed JDAF under different cross-domain scenarios and provides insightful analyses. Section V concludes the article.

II. Related Works

A. Semantic Segmentation

Semantic segmentation of HRS images is a key technique for various applications, such as precision agriculture, disaster prediction, urban traffic management and planning, etc. The advancement of convolutional neural networks (CNNs) has greatly impacted the field of semantic segmentation in computer vision research [6], [7], [8], [9], [10], [11], [12], [13], [14], [19]. The fully convolutional network (FCN) [8] was the first approach to use CNNs for semantic segmentation, and since then, FCN-based methods have made significant progress in the semantic segmentation of images. Several improvements have been made to the original FCN architecture. Chen et al. [9] and Yu and Koltun [10] remove the last two downsampling layers to obtain dense prediction and use dilated convolutions to increase the receptive field. Other architectures, such as Unet [12], Deeplab-v3+ [13], and RefineNet [14], adopt encoder–decoder structures that combine the information from low- to high-level layers to predict segmentation masks. BiSeNet [19] proposes a bilateral segmentation path to generate high-resolution features and obtain a sufficient receptive field.

B. Domain Adaptation for Semantic Segmentation

UDA aims to transfer knowledge from a labeled source domain to an unlabeled target domain and has been well-studied in image recognition tasks [15], [16], [17], [18]. These approaches, however, perform poorly in more challenging and complex tasks, such as semantic segmentation. To solve this problem, a group of self-supervised learning adaptive models [37], [38], [39], [40], [41] are proposed by using pseudo labels to implicitly learn domain-invariant representations. For example, SEANet [39] exploits

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a self-ensembling attention network to combine the benefits of both the ensemble model and self-training to reduce the domain gaps and improve the feature transferability of the semantic segmentation network. ProDA [41] uses the feature centroids of classes to estimate the likelihood of pseudo labels and aligns the prototypical assignments based on relative feature distances.

Recently, most existing methods (i.e., GAN-based methods) maintain semantic consistency through adversarial training between the network and discriminator [20], [21], [22], [23], [24], [25], [26], [27], [28]. According to the different alignment spaces, the GAN-based methods can be roughly divided into three categories: input space alignment, feature space alignment, and output space alignment. CycleGAN [22] and DualGAN [23] are the most representative methods for **input space alignment** that allow image styles to be converted from the source domain to the target domain. These methods, however, fail to align semantic features across domains, resulting in poor-quality results. As for the methods of **feature space alignment**, Advent [26] and IntraDA [28] are the primary methods for aligning features across multiple feature spaces using GANs and just modeling marginal distributions across domains. These methods, however, ignore category labels and do not truly align joint distributions, which can lead to some of the categories that were initially aligned between the source and target domains being mismapped, resulting in poor segmentation results in the target domain, as shown in [25]. Among the methods of **output space alignment**, AdaptSegNet [24] focuses on adversarial learning training at the output level to reduce interdomain feature differences from global spatial relationships. Because the output space often suffers a significant loss of information compared to the feature space, this means that forcing only the consistency of the output space distributions does not ensure that the marginal distributions between the two data domains are properly aligned.

In a word, the above GAN-based methods essentially only explicitly model alignment of marginal distributions. Unlike the previous methods, our JDAF explicitly models not only the alignment of marginal distributions but also the alignment of conditional distributions, which ensures precise alignment of joint distributions between the source and target domains.

Although the UDA methods described above achieve impressive results in the semantic segmentation of images, they are mainly applied to natural images; however, as previously described, HRS images differ from natural images in several aspects, which causes spatial inconsistency in segmentation maps when applying segmentation networks to HRS images.

C. Domain Adaptation for Semantic Segmentation in Remote Sensing Images

UDA has been proposed for HRS images to address data bias [29], [30], [31], [32], [33]. GAN-RSDA [31] and DualGAN [32] use adversarial learning to achieve GAN-based feature transferability and image-level style transfer networks, respectively. They optimize their networks by dynamically adjusting constraint weights. CCDA [33] introduces an adaptation strategy from easy to difficult, which combines self-supervised learning and adversarial learning to adapt from local to global; however, the presence of noise in the pseudo labeling of self-supervised learning affects the migration effect of the model.

These existing methods implicitly model marginal distribution and conditional distribution between different domains; however, the complex acquisition scenarios for HRS images usually cause inconsistent discriminative category decision boundaries between the two domains; therefore, the alignment of conditional distributions for cross-domain semantic segmentation of HRS images requires further investigation. Meanwhile, the existing methods with self-supervision may cause the model to learn erroneous knowledge from the pseudo labels and reduce segmentation accuracy.

III. Proposed Method

Our work focuses on how to migrate the source domain model to the target domain from the perspective of alignment of joint distributions. We first give the definition of the joint distribution alignment problem. Supposing that $D_S = \{(x_s, y_s)\}_{s\in S}$ and $D_T = \{(x_t)\}_{t\in T}$ denote source domain and target domain, respectively, I is the feature representation where $I = F(x)$ and F is a learned backbone network for feature extraction, and x and y denote the image and category, respectively.

Generally, the source domain data D_S and the target domain data D_T are obtained by sampling from different joint probability distributions $P_s(y_s, I_s)$ and $P_t(y_t, I_t)$, respectively, and usually $P_s(y_s, I_s) \neq P_t(y_t, I_t)$. The purpose of UDA is to learn a feature extractor F that can generalize to both domains and achieve $P_s(y_s, I_s) = P_t(y_t, I_t)$.

The joint distribution of the data can be written from the Bayesian formula:

$$P(y, I) = P(I|y)P(y) = P(y|I)P(I). \tag{1}$$

To align joint distributions $P_s(y_s, I_s) = P_t(y_t, I_t)$, we have two ways: aligning $P(I|y)P(y)$ or $P(y|I)P(I)$. For the former, since the annotation distributions $P(y)$ is actually inconsistent between domains and unable to be aligned, it may be difficult to align the joint distributions provided that only $P(I|y)$ is aligned; therefore, $P(y|I)$ and $P(I)$ of the source and target domains are chosen to be aligned separately to achieve the joint distribution alignment of the source and target domains.

Our work, moreover, aims at transferring a pretrained semantic segmentation model from a HRS image dataset to a new unlabeled dataset with high performance of segmentation. Given image x_s from the source domain and the feature extractor (backbone) F, the purpose of model training for semantic segmentation is to optimize the backbone network F so that the segmentation loss L_{seg} is minimized:

$$L_{\text{seg}} = -\sum_{u=1}^{U}\sum_{v=1}^{V}\sum_{c=1}^{C_K} Y^{(u,v,c)} \log(\text{cls}(I)) \tag{2}$$

where U and V represent the feature sizes of output space, respectively, the data has a total of C_K categories, and cls denotes a pixel-level classifier.

As shown in Fig. 1, we propose the JDAF that fully aligns the joint distributions of the source and target domains, enabling cross-domain adaptation of the model.

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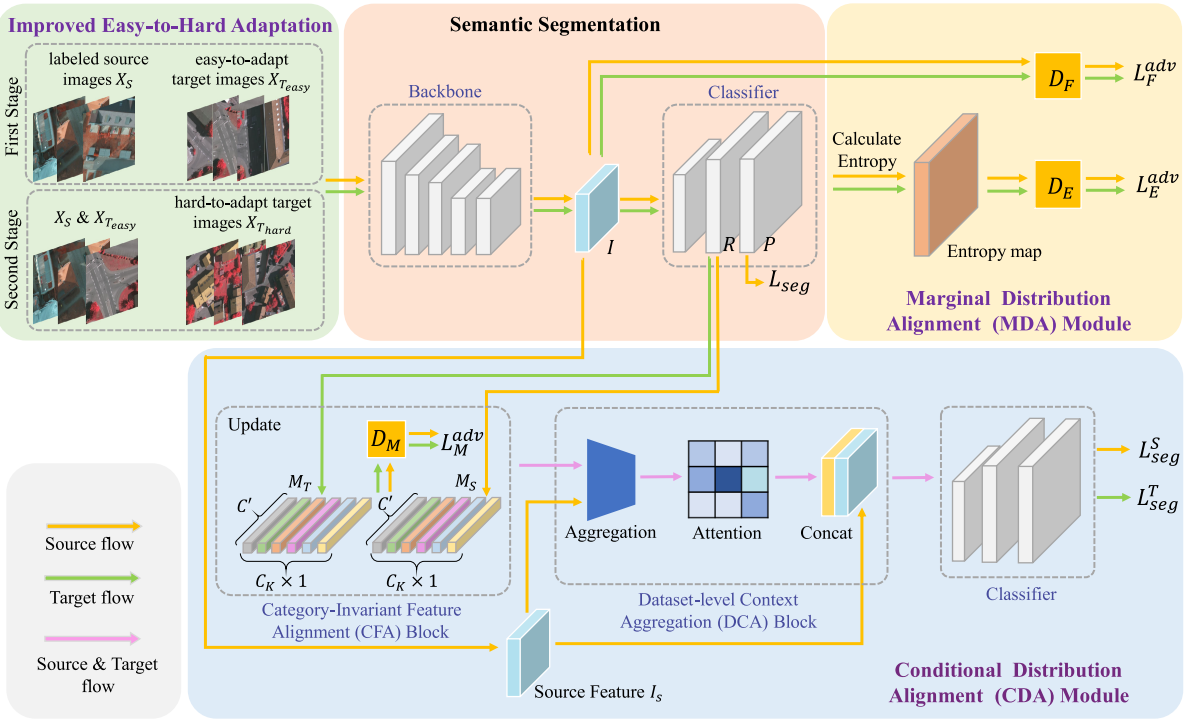


Fig. 1. Structure of the proposed JDAF. The adaptation process is divided into two phases (easy and difficult adaptation phases) after the target data is divided into easy-to-adapt and hard-to-adapt samples. The model performs joint distribution alignment (including MDA and CDA) for both easy-to-adapt and hard-to-adapt samples. In particular, in the second stage of the adaptation process, the easy-to-adapt samples and their pseudo labels predicted by the model are grouped into the source domain samples; moreover, in the second stage, the proposed UAL method is applied to both MDA and CDA.

It is structurally mainly composed of two components: the MDA module and the CDA module; moreover, inspired by CCDA [33], we improve the two-stage adaptation strategy from easy to difficult by means of the proposed UAL method, which is designed to filter the noise in the pseudo labels and better adapt hard target images at the pixel level by assigning larger weights to harder images during alignment.

A. MDA Module

Usually, data from source and target domains suffers from large local semantic feature differences and global spatial differences, which tend to cause the image representations of the target domain $I_t = F(x_t)$ to marginally deviate from the data distribution of the source domain, i.e., $P_s(I_s) \neq P_t(I_t)$. The local semantic feature differences mean the texture and appearance differences of objects in different domains caused by variations in imaging sensors, lighting, atmospheric conditions, etc. The global spatial differences mean the structural layout differences of objects in different scenes caused by differences in geographic locations, land cover types, urbanization levels, etc. To solve such issues, an adversarial learning method is adopted to align the marginal distributions between the two domains in our proposed framework, inspired by CCDA [33].

1) Local Semantic Feature Alignment: First, a semantic-level domain discriminator D_F, which adopts a spatially dense domain discriminator, is used to adapt features between domains, preserving the spatial resolution of the features. The discriminator D_F consists of five 4×4 convolution layers with a stride of 2, and the channel number of each layer is [128, 256, 512, 1024, 1]. The local semantic consistency between domains is maintained by adversarial training between the segmentation network and the discriminator D_F from the semantic level. The adversarial loss of D_F can be written as:

$$L_F^{\text{adv}}(D_F) = -\frac{1}{HW}\sum_{h,w} d \log\left(D_F\left(I_s^{(h,w)}\right)\right) - \frac{1}{HW}\sum_{h,w}(1-d)\log\left(1 - D_F\left(I_t^{(h,w)}\right)\right). \quad (3)$$

The discriminator D_F classifies the image into the source or target domain from the semantic feature level, using the domain labels $d = 1$ and $d = 0$ for the images of source and target domains, respectively. $I^{(h,w)}$ is the feature representation of the feature map located at the spatial location (h, w). The H and W are the sizes of the feature map.

2) Global Spatial Feature Alignment: Since the entropy map obtained from the output-level feature map encodes the global layout information of all semantic categories, similar to a kind of edge detection results, an entropy-level domain-based discriminator D_E with a structure similar to the discriminator D_F is adopted to measure the cross-domain variation of global spatial information. Entropy-level domain adaptation is performed on the prediction layer of category probability $I_p = C_1(I) \in \mathbb{R}^{U \times V \times C_K}$. The adversarial loss of discriminator D_E can be written as:

$$L_E^{\text{adv}}(D_E) = -\frac{1}{UV}\sum_{u,v} d \log\left(D_E\left(\varepsilon\left(I_{s,p}^{(u,v)}\right)\right)\right) - \frac{1}{UV}\sum_{u,v}(1-d)\log\left(1 - D_E\left(\varepsilon\left(I_{t,p}^{(u,v)}\right)\right)\right) \quad (4)$$

where $I_{s,p}^{(u,v)}$ and $I_{t,p}^{(u,v)}$ represent the probability predictions of the model for the data of source and target domains at the spatial location (u, v), respectively. The $\varepsilon(\cdot)$ converts the class-specific output representation into an entropy value

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map. The entropy-level domain discriminator D_E is similar to the semantic-level domain discriminator D_F, distinguishing whether the entropy map is from the source or target domain.

MDA is an unsupervised cross-domain feature matching method that does not require pixel-level annotations; however, it does not account for the category decision boundaries, which may result in uneven and biased adaptation and degrade the discriminative power for the target domain. The MDA, moreover, is used alone for semantic segmentation, ignoring cross-domain transferability changes across different categories, which may result in negative shifts, i.e., over-adaptation of some nontransferable categories. Thus, the CDA module is further proposed to solve the above problem by considering category transferability.

B. CDA Module

Based on the previous analysis, we need to align the cross-domain conditional distributions $P_s(y_s|I_s) = P_t(y_t|I_t)$. A feasible solution is to use the source domain model to predict the target domain category. The conditional probabilities on the target domain can then be transformed as follows:

$$P_t(y_t|I_t) = \frac{P_t(I_t|y_t)P_t(y_t)}{P_t(I_t)} \approx \frac{P_t(I_t|\hat{y}_t)P_t(\hat{y}_t)}{P_t(I_t)}. \tag{5}$$

The conditional probability transformation on the source domain resembles (5); moreover, considering $P_t(\hat{y}_t) = P_s(\hat{y}_s)$, the pseudo-category label of the target domain $\hat{y}_t$ are predicted from the source domain model, and it should have the same probability distribution as $\hat{y}_s$ for the source domain, and $P_s(I_s) = P_t(I_t)$ is achieved by MDA. The rest is that we need to extract and align $P(I|y)$ so as to achieve the alignment of $P(y|I)$.

To align the cross-domain conditional distributions, The CDA module extracts and adapts the category-invariant features of the source and target domains. It consists of three components: a CFA block, a DCA block, and a classifier. The backbone network F and these components form a new segmentation network.

1) CFA Block: Extracting and aligning cross-domain category features is essential for CDA; however, the existing UDA semantic segmentation methods based on adversarial learning in each iteration only propagate gradients from the adversarial loss of the current batch images. As a result, the extracted features only contain the representation of the current inputs (considered as pseudo-category-invariant features), which are insufficient to represent the invariant features of domains. Aligning the pseudo-category-invariant features achieves only the biased alignment of conditional distributions, resulting in poor knowledge or negative transfer from the source to the target domain, where some adapted categories may shift during training. Hence, our goal is to design a CFA block to extract and align the cross-domain category-invariant features for an unbiased alignment.

As shown in Fig. 1, the CFA block maintains invariant domain-level representations M_S and M_T in a category format, which allows the network to capture the information of invariant features of the whole domain. M is a data vector with size $C' \times C_K \times 1$, where C' is the vector size of each category and C_K is the number of categories.

During training, we first randomly select a pixel representation for each category from the train set to initialize M. In each batch of training, the CFA block extracts the pseudo-category-invariant features R_{c_k} of the source domain $D_S = \{(x_s, y_s)\}_{s\in S}$ and target domain $D_T = \{(x_t)\}_{t\in T}$ to update the M_S and M_T, respectively. Since the target domain lacks labels, the source and target domains have different ways for extracting the pseudo-category-invariant features.

For the source domain, the set of pseudo features R_{s,c_k} of all pixels in category $c_k \in C_K$ is extracted from the category output space R:

$$R_{c_k} = \left\{ R_{[i]}^{UV\times C'} | \mathrm{GT}_{[i]} = c_k \right\} \in \mathbb{R}^{N_{c_k}\times C'} \tag{6}$$

where R_{c_k} is collected from the representation of pixel i of the category output space R, pixel i is the index of the pixels of the c_k category in $\mathrm{GT} \in \mathbb{R}^{UV}$, GT is the ground truth with category labels, and N_{c_k} is the number of pixels of category c_k in R.

For the target domain, due to the lack of ground truth for determining the pixel category, the predictions of the model's outputs on the target domain images are used as pseudo labels $P_t = \{p_{t,c_k}^{(u,v)} \in \mathbb{R}^{U\times V\times C_K}\}_{t\in T}$ to obtain the set of pseudo features R_{t,c_k}, where $p_{t,c_k}^{(u,v)}$ denotes the probability that pixel (u, v) belongs to category c_k.

By calculating the maximum value of $p_t^{(u,v)}$ ($\mu = \max(p_{t,c_k}^{(u,v)})$), the pseudo label is determined, where the index of μ is specified as the category of p_t at pixel (u, v); thus, the matrix $\hat{y}_t = \mathrm{index}(\mu) \in \mathbb{R}^{U\times V}$ of the predicted categorical labels can be obtained.

Aiming to obtain the high-quality pseudo labels, the entropy map of $p_t^{(u,v)}$ is used to retain the high-confidence pixels and discard the low-confidence pixels. A threshold σ is used to determine the set $\hat{y}_t'$ of pixels with high confidence in the pseudo labels:

$$\hat{y}_t' = \left\{ \mathbb{R}^{U\times V} | E^{(u,v)} \le \sigma \right\} \tag{7}$$

where $E^{(u,v)}$ represents the entropy value at the pixel (u, v).

The pseudo labels with entropy values greater than the threshold σ are discarded, while others are retained. The set of target pseudo feature R_{t,c_k} can be obtained as:

$$R_{t,c_k} = \left\{ R_{[i]}^{UV\times C'} | \hat{y}_{t,[i]}' = c_k \right\} \in \mathbb{R}^{N_{c_k}\times C'}. \tag{8}$$

Then, the pseudo-category-invariant features R_{s,c_k} and R_{t,c_k} are employed to update the M_S and M_T by using the same way. Here, the CFA block uses cosine similarity to calculate the similarity between R_{c_k} and M_{c_k} to update features

$$S_{c_k} = \frac{R_{c_k} \cdot M_{c_k}}{\left\|R_{c_k}\right\|_2 \cdot \left\|M_{c_k}\right\|_2} \tag{9}$$

where $M_{c_k} \in \mathbb{R}^{1\times C'}$, $S_{c_k} \in \mathbb{R}^{1\times N_{c_k}}$, and $\|\cdot\|_2$ represents L_2 norm. The feature representation of category c_k in the current batch is calculated by similarity weighting:

$$R_{c_k}^{i\,\prime} = \sum_{i=1}^{N_{c_k}} \frac{1 - S_{c_k}^i}{\sum_{j=1}^{N_{c_k}} \left(1 - S_{c_k}^j\right)} \cdot R_{c_k}^i, \quad i \in N_{c_k}. \tag{10}$$

The category-invariant features for each category in the source and target domains are further updated by applying

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the moving average:

$$M'_{S,c_k} = (1-m)\cdot M_{S,c_k} + m\cdot R'_{s,c_k} \tag{11}$$

$$M'_{T,c_k} = (1-m)\cdot M_{T,c_k} + m\cdot R'_{t,c_k} \tag{12}$$

where M'_{S,c_k} and M'_{T,c_k} are the invariant-feature representations of updated category c_k for the source and target domains, respectively, and m is the momentum. A polynomial annealing method is used to update m:

$$m_t = \left(1-\frac{t}{T}\right)^p \cdot \left(m_0 - \frac{m_0}{100}\right) + \frac{m_0}{100}, \quad t \in [0,T] \tag{13}$$

where t and T are the current and total number of training iterations, respectively.

To adapt category-invariant features, a discriminator D_M is adopted to measure the discrepancy between the conditional distributions of the source and target domains by using adversarial training. The discriminator D_M consists of three linear layers with output dimensions of [1024, 10, 1]. The adversarial loss of the discriminator D_M can be written as:

$$L_M^{\text{adv}}(D_M) = -\frac{1}{C_K C'}\sum d\log(D_M(\xi(M_S))) - \frac{1}{C_K C'}\sum (1-d)\log(1-D_M(\xi(M_T))) \tag{14}$$

where $\xi(\cdot)$ denotes the reshape operation and $\xi(M) \in \mathbb{R}^{1\times C_K C'}$. Like the semantic-level discriminator D_F and the entropy-level discriminator D_E, the discriminator D_M also distinguishes the category-invariant features from the source or target domains.

2) DCA Block: The M_S and M_T capture the dataset-level contextual information of their corresponding domains, which contain rich contextual information and can enhance the accuracy of the category probability prediction. As shown in Fig. 2, the block is designed to enhance the feature $I' = \{I_{s,s}, I_{s,t}\} \in \mathbb{R}^{C'\times H\times W}$ by aggregating the $M = \{M_S, M_T\} \in \mathbb{R}^{C'\times C_K\times 1}$ with the source feature $I_s \in \mathbb{R}^{C'\times H\times W}$ extracted from the backbone network F. One of the sets of features $M = \{M_S, M_T\}$ is selected to use two 1×1 convolutional layers, one to generate feature $Q \in \mathbb{R}^{C'\times C_K\times 1}$ and the other to generate feature $K \in \mathbb{R}^{C'\times C_K\times 1}$. The source feature I_s uses a 1×1 convolutional layer to generate the feature $V \in \mathbb{R}^{C'\times H\times W}$. To reduce the model parameters, the source and target domains share these three 1×1 convolutional layers. Subsequently, the reconstructed features K and V are matrix multiplied, and then the softmax function is used to calculate the affinity attention map S between M_S (or M_T) and the extracted feature I_s:

$$s_{i,j} = \frac{\exp(K_i \cdot V_j)}{\sum_{i=1}^{C_K}\exp(K_i \cdot V_j)} \tag{15}$$

where $s_{i,j} \in S$ measures the correlation between the ith category of feature K and jth channel of feature V, $i = [1,2,\ldots,C_K]$, $j = [1,2,\ldots,C]$. The attention map S' is obtained by taking a weighted sum of the affinity attention map S and feature Q. The feature I_s is then concatenated with the attention map S' to obtain the augmented features:

$$I' = \theta(\text{concat}(\varphi(S'),\ I_s)) \tag{16}$$

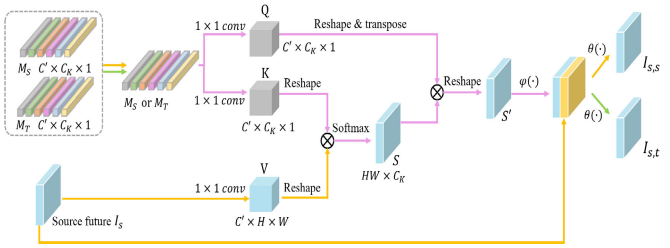


Fig. 2. Detailed structure of DCA block.

where $\theta(\cdot)$ and $\varphi(\cdot)$ represent the 1×1 convolution operation, and the source and target domains share these two convolutional layers. The enhanced feature I' is adaptively enhanced according to the correlation between the category-invariant features M and I_s. As such, the contextual information can be better captured and the segmentation accuracy can be improved.

When M_T is sufficiently adapted to the source feature M_S, the semantic information contained in M_T is close to that of M_S; therefore, the M_T can replace the M_S and provide the contextual information and category feature for the source feature I_s to obtain the enhanced feature I'. Here, the following two aggregation ways can be employed to adaptively aggregate dataset-level contextual information for the corresponding domain:

$$I_{s,s} = \text{DCA}(I_s,\ M_S) \tag{17}$$

$$I_{s,t} = \text{DCA}(I_s,\ M_T) \tag{18}$$

where DCA represents all operations of DCA block. The $I_{s,s}$ and $I_{s,t}$ denote the enhanced features after adaptive aggregation of the source features I_s and M_S, and I_s and M_T, respectively.

The enhanced features $I_{s,s}$ and $I_{s,t}$ output the category probability prediction after the classifier. The model optimizes the segmentation network by calculating the segmentation loss under the supervision of the pixel-level labels of the source domain. The cross-entropy loss function is used as the segmentation loss in this work:

$$L_{\text{seg}}^S(F,A) = -\sum_{u=1}^{U}\sum_{v=1}^{V}\sum_{c=1}^{C} Y_s^{(u,v,c)}\log\left(\text{cls}\left(I_{s,s}\right)\right) \tag{19}$$

$$L_{\text{seg}}^T(F,\ A) = -\sum_{u=1}^{U}\sum_{v=1}^{V}\sum_{c=1}^{C} Y_s^{(u,v,c)}\log\left(\text{cls}\left(I_{s,t}\right)\right) \tag{20}$$

where cls denotes a classifier and cls$(\cdot)$ is capable of converting high-level representations into class-specific output spaces. The category probability prediction for each pixel from $I_{s,t}$ is also supervised by the ground truth of source domain, which can help M_S adapt better to M_T, and align the interdomain conditional distributions further.

C. Improved Easy-to-Hard Adaptation Strategy

Inspired by the superior performance achieved by the easy-to-hard adaptation strategy [28], [33], our framework also employs this adaptation strategy. The existing methods based on the easy-to-hard adaptation strategy, however, have important defects in ignoring the noise present in the pseudo labels.

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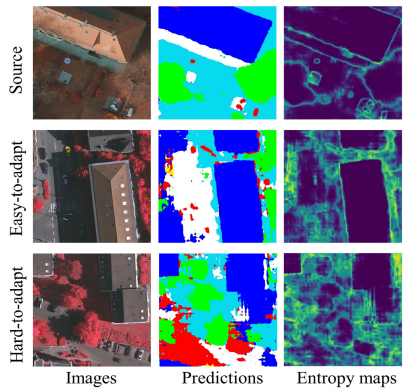


Fig. 3. Predictions and corresponding entropy maps for the semantic segmentation model of the source domain on an image from the source domain, an easy-to-adapt image, and a hard-to-adapt image from the target domain.

This problem significantly affects the adaptation effect of the model because the model may learn the wrong knowledge from the pseudo labels; therefore, we propose the UAL method to improve this easy-to-hard adaptation strategy.

1) Division of Target Domain Samples: As shown in Fig. 3, the semantic segmentation model trained only on the source domain usually produces over-confident (low entropy) prediction for an image from the source domain (the first row) and under-confident (high entropy) predictions for the images from the target domain (the second and last rows); moreover, there are intradomain differences among the images from the target domain (the second and last rows), with some images having cleaner entropy maps and some images having more noisy ones. Considering that the cross-domain adaptation for all the patches from the target domain should follow the two-stage adaptation, the easy-to-adapt patches should be adapted before the difficult ones.

In this article, we adopt the mean value of the entropy map as a cross-domain adaptation uncertainty score [33]:

$$\overline{E} = \frac{1}{UVC_K} \sum_{u=1}^{U} \sum_{v=1}^{V} \sum_{c=1}^{C_K} (-\psi(P) \log(\psi(P))) \tag{21}$$

where ψ represents the softmax activation function and P is the category probability feature map of the output layer. A lower score means a cleaner entropy map and an easy-to-adapt sample, and vice versa; thus, all the images from the target domain can be ranked by the cross-domain adaptation uncertainty scores.

After the easy-to-adapt samples $D_{T_{\text{easy}}}$ (including the images and the corresponding pseudo labels) are adapted to the source domain, they are merged into the source domain. The new source domain $D_{S_{\text{new}}}$ can be obtained by adding $D_{T_{\text{easy}}}$ to the source domain. In the second stage, the model will focus on adapting the hard-to-adapt samples.

2) UAL Method: To effectively filter out false knowledge from low-confidence pixels and extract useful features from pseudo labels, we propose the UAL method (including self-supervised and adversarial learning), which can be applied to $\hat{y}_t \in D_{S_{\text{new}}}$ to help our architecture learn better domain-invariant features.

a) Self-supervised learning: Since the easy-to-adapt samples have pseudo labels $\hat{y}_t$, they may differ from the ground truth. Some pixels have low-confidence pseudo labels and are more likely to be different from the ground truth. To better use the pseudo labels of easy-to-adapt samples, consider that the pixels with the high-confidence pseudo labels need to contribute more to loss. In contrast, the pixels of the low-confidence pseudo labels may introduce false knowledge and impair model adaptation, and thus, they should contribute less to the loss. By means of the standard self-supervised learning approaches [28], [33], [38], a threshold can be set to select the pseudo labels with high confidence; however, it may be very difficult to choose an appropriate threshold for the above methods because too large thresholds would leave little available target information and too small thresholds would produce too many noisy labels, thus impacting adaptation performance.

To solve the above problem, an uncertainty-adaptive self-supervised loss is proposed. In the second stage, the pixel-level weighting is performed for two target predictions supervised in $D_{T_{\text{easy}}}$ samples as:

$$L_{\text{seg}}^{i} = -\sum_{u=1,\, v=1}^{U,\, V} \lambda_1 \left(1 - E_t^{(u,v)}\right)^2 \sum_{c=1}^{C_K} \widehat{Y}_t^{(u,\, v,\, c)} \log\left(\text{cls}\left(I_p\right)\right) \tag{22}$$

where $i \in \{1, 2\}$ and I_p is the feature representation. $i = 1$ represents the segmentation loss of samples $D_{T_{\text{easy}}}$ in the original segmentation network and $i = 2$ denotes the segmentation loss of samples $D_{T_{\text{easy}}}$ after aggregation of the source domain feature I_s with M_T. The classifier is denoted as cls. The λ_1 is a correction factor for the uncertain-adaptive self-supervised loss.

b) Adversarial learning: To further adapt the low-confidence samples among the easy-to-adapt samples and make them closer to the source domain samples, consider that the low-confidence samples need to contribute more to loss to enhance the model's learning for these samples; thus, the sum of the pixel entropy values are used to measure the difficulty of each target image. In the second stage, an uncertainty-adaptive adversarial loss is proposed:

$$L_j^{\text{adv}} = -\sum_{u,v} \lambda_2 \left(1 + \left(\sum_{u,\, v} E_t^{(u,\, v)}\right)^2\right) \cdot d \log\left(D\left(I_{\text{easy}}\right)\right) - \sum_{u,v} (1 - d) \log(1 - D(I_{\text{hard}})) \tag{23}$$

where $j \in \{1, 2\}$. $j = 1$ denotes the semantic-level adversarial loss and $j = 2$ means the entropy-level discriminator loss. I_{easy} and I_{hard} represent the features of easy-to-adapt and hard-to-adapt samples, respectively, and λ_2 is a correction factor for the uncertainty-adaptive adversarial loss.

3) Overall Loss Function: The overall loss function of the proposed method can be written as follows:

$$\max_{D_F} \min_{F} L_{\text{seg}}(F) - \mu L_F^{\text{adv}}(D_F) \tag{24}$$

$$\max_{D_E} \min_{F} L_{\text{seg}}(F) - \mu L_E^{\text{adv}}(D_E) \tag{25}$$

$$\max_{D_M} \min_{F,A} L_{\text{seg}}(F) + L_{\text{seg}}^{S}(F, A) + L_{\text{seg}}^{T}(F, A) - \mu L_M^{\text{adv}}(D_M) \tag{26}$$

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where μ denotes the hyperparameter of the discriminability of the model for the source domain data and the degree of transfer for the target domain data. The semanticlevel, the entropy-level, and the category-invariant feature-level discriminators can be trained adversarially with the backbone according to (24)–(26), respectively.

4) Improved Easy-to-Hard Adaptation Strategy: The samples of the target domain are divided into two groups, the easy-to-adapt samples and the hard-to-adapt samples, for the two-stage training according to the division of the samples of the target domain (i.e., entropy sorting).

a) First stage: The MDA module aligns the source domain samples with the easy-to-adapt samples using the semantic-level and entropy-level discriminators from local to global so as to realize the preliminary alignment of marginal distributions. The CDA module extracts the category-invariant features and aligns the conditional distributions between two domains so as to solve the biased adaptation problem caused by ignoring the class decision boundaries.

b) Second stage: The model focuses on the joint distribution alignment for the hard-to-adapt samples. The easy-to-adapt samples and pseudo labels are grouped into the source domain to obtain $D_{S_{\text{new}}}$. By using the UAL method, the MDA and CDA modules improve the use of knowledge in the pseudo labels to adapt the hard-to-adapt samples.

IV. Experimental Results and Analyses

A. Dataset Description

Following common UDA protocols for HRS images [29], [31], [32], [33], we use two commonly used spaceflight datasets, including the Potsdam and Vaihingen datasets constructed by the International Society for Photogrammetry and Remote Sensing (ISPRS) [42].

1) Potsdam: The Potsdam dataset consists of 38 very high-resolution (VHR) true orthophotos (TOP) images of 6000 × 6000 pixels with a spatial resolution of 5 cm. The Potsdam dataset has different imaging modes, including the IR-R-G channel, R-G-B channel, and R-G-B-IR channel. To follow the experimental setup in [32], we performed the following experiments using the first two imaging modes (IR-R-G/R-G-B).

2) Vaihingen: The Vaihingen dataset consists of 33 VHR TOP images with an average size of 2494 × 2064 pixels and a spatial resolution of 9 cm. The imaging channel is IR-R-G channel.

Because of GPU memory limitations, the images of the above two datasets are cropped to 512 × 512 pixel patches to train the segmentation network for UDA properly, which is consistent with the experimental setting in [29], [31], [32], [33]. Thus, the cropping yielded 4598 Potsdam patches and 1696 Vaihingen patches.

In our experiments, we set up four UDA semantic segmentation tasks.

3) PIRRG_to_V Task: The Potsdam dataset with the IR-R-G channel is used as the source domain dataset for segmentation training, and the Vaihingen dataset with the IR-R-G channel is used as the target domain dataset.

4) PRGB_to_V Task: The Potsdam dataset with the R-G-B channel is taken as the source domain dataset for segmentation training, and the Vaihingen dataset with the IR-R-G channel is used as the target domain dataset.

5) V_to_PIRRG Task: The Vaihingen dataset with IR-R-G channel is used as the source domain dataset for segmentation training, and the Potsdam dataset with the IR-R-G channel is adopted as the target domain dataset.

6) V_to_PRGB Task: The Vaihingen dataset with the IR-R-G channel as the source domain dataset for segmentation training and the Potsdam dataset with the R-G-B channel as the target domain dataset.

We apply Deeplab-v3+ [13] as the baseline segmentation model. The proposed JDAF is deployed on the baseline segmentation model and evaluated on the above four tasks to validate the superiority of our approach.

B. Implementation Details

To learn sufficient transferable knowledge for the subsequent domain adaptation process, we first make the baseline model fully trained on the source domain data. We use the PyTorch toolbox to implement our network on a single RTX 3090 GPU with 24 GB of memory. The baseline model is trained on the source domain using a stochastic gradient descent (SGD) optimizer with an initial learning rate set to 0.0005, a momentum of 0.9, a weight decay of 0.0005, a batch size of 4, and decay according to a polynomial annealing algorithm [9].

Our JDAF is then deployed on the baseline model to perform a cross-domain adaptation for HRS images. For the semantic segmentation network, training details are the same as the training of the baseline model. For all discriminator networks, training is performed using the Adam optimizer with an initial learning rate of 0.0001 and momentum of 0.9 and 0.99. The proportion of pixels with high confidence is set to 0.3, the hyperparameters λ_1, λ_2, and μ to 5, 2, and 0.1, respectively, and the length of each class of domain category-invariant feature vector is set to 1024. The ratio of easy-to-adapt samples to total target samples is set to 0.5 according to CCDA [33]. The batch size of source and target domain data is set to 4 and decayed according to the polynomial annealing procedure.

C. Experimental Results

We are devoted to studying the impact of a two-stage adaptation strategy on common segmentation networks and designing several models, including baseline with MDA (denoting the original baseline model with the MDA module), adapted baseline with CDA (denoting the marginal distribution fully aligned baseline model with the CDA module), baseline with MDA and UAL (denoting the adapted best model of the first stage with the MDA module and the UAL method), and adapted baseline with CDA and UAL (denoting the first stage model with the CDA module and the UAL method after the features of the marginal distribution are fully aligned).

We conduct experiments in two cross-domain scenarios: 1) only geographic locations are dynamically changing and

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TABLE I
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT UDA MODELS ON THE PIRRG_TO_V TASK (%)

Method	Clutter/ background		Impervious surfaces		Car		Tree		Low vegetation		Building		Overall	
	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	mIOU	F1-Score
Baseline (Deeplab-V3+) [13] (2018)	5.71	10.79	35.84	52.73	20.27	33.70	54.95	70.92	17.88	30.26	51.59	68.06	31.04	44.40
BiSeNet only [19] (2018)	1.67	3.28	38.46	55.55	9.42	17.20	45.54	62.57	23.04	37.44	28.15	43.93	24.38	36.65
SEANet [39] (2019)	11.11	20.00	45.57	62.59	32.71	49.29	50.42	66.83	23.06	37.45	57.99	73.27	36.81	51.60
AdaptSegNet [24] (2018)	4.60	8.76	54.39	70.39	6.40	11.99	52.65	68.96	28.98	44.91	63.14	77.40	35.02	47.05
GAN-RSDA [31] (2019)	2.12	4.15	39.88	57.02	8.20	15.15	26.56	41.97	26.53	41.94	40.97	58.10	24.04	36.40
ProDA [41] (2021)	3.99	8.21	62.51	76.85	39.20	56.52	56.26	72.09	34.49	51.65	71.61	82.95	44.68	58.05
DualGAN [32] (2021)	29.66	45.65	49.41	66.13	34.34	51.09	57.66	73.14	38.87	55.97	62.30	76.77	45.38	61.43
CCDA [33] (2022)	20.71	31.34	67.74	80.13	**44.90**	**61.94**	55.03	71.90	47.02	64.16	76.75	86.65	52.03	66.02
Adaptation from Labeled Source Patches to Easy-to-adapt Target Patches														
Baseline+MDA (ours)	10.34	18.74	65.71	79.31	40.62	57.77	56.01	71.81	40.95	58.11	77.28	87.18	48.49	62.15
Adapted-Baseline+CDA (ours)	23.58	38.17	**68.78**	**81.50**	41.95	59.10	58.28	73.64	43.17	60.31	**78.72**	**88.09**	52.41	66.80
Adaptation from Labeled Source Patches and Easy-to-adapt Target Patches to Hard-to-adapt Target Patches														
Baseline+MDA+UAL (ours)	35.56	52.46	68.66	81.42	43.56	60.69	**58.61**	**73.90**	45.50	62.54	76.81	86.88	54.78	69.65
Adapted-Baseline+CDA+UAL (ours)	**38.65**	**55.75**	68.76	81.49	42.76	59.90	58.38	73.72	**47.39**	**64.30**	77.19	87.13	**55.52**	**70.38**

2) both geographic locations and imaging modes are dynamically changing.

1) Dynamically Changing From Geographic Locations: The quantitative adaptation results on the PIRRG_to_V task are presented in Table I. This cross-domain scenario has domain differences only in geographic location, spatial resolution, and illumination. On Deeplab V3+, our adaptation result reached 55.52%, and five categories are optimal. As we discussed before, JDAF aims to align the joint distribution features between domains. Aligning the conditional distributions after MDA can reduce the negative transfer of outlier categories. Compared with the previous works [19], [24], [31], [32], [33], [39], [41], our method has made a significant improvement in outlier categories of PIRRG_to_V task, such as clutter/background, tree, and low vegetation. The CDA improves the segmentation accuracy of each category without sacrificing the accuracy of one category for the others. This matches our previous analysis (the alignment of the discriminative classification decision surfaces within two domains), which confirms the CDA module's effectiveness. JDAF, however, performs worse than CCDA in the car category, which may be due to the fact that the car category only accounts for 1.7% and 1.3% of the pixels in the Potsdam and Vaihingen datasets, respectively, making it difficult for our model to fully extract the category-invariant features of the car from the source and target domains. Although JDAF is inferior to CCDA in the car category with a small percentage of pixels, the overall performance of our JDAF is still significantly better than CCDA on the two datasets.

We pay more attention to comparisons with the latest category-level domain adaptation work [32], [33]. These two methods show remarkable improvement in common categories in Table I, and especially, an easy-to-hard adaptation strategy and pseudo labels are used to learn latent feature alignment [33]. Our method does pixel-level reweighting of supervision based on the pseudo labels' pixel confidence to avoid aligning misaligned features. The proposed UAL method further improves the cross-domain segmentation accuracy by 2.86%.

Our work is inspired by MCIBI [43]. Like MCIBI, we use attention-based adaptive aggregation of category-invariant features to enhance the current feature representation and achieve more accurate segmentation predictions. We differ from MCIBI in that we extract the category-invariant features from both the source and target domains. Meanwhile, the target domain category-invariant feature M_T replace M_S for the adaptive aggregation of the source domain feature, and the prediction results are supervised by the source domain labels. This helps further align the conditional distributions, due to that M_T can be gradually aligned with M_S under supervision.

On the other hand, the above experiments are performed for the PIRRG_to_V task. In order to get a comprehensive view of the impact of different cross-domain scenarios on the proposed method, we swap the source and target domains and perform the V_to_PIRRG task. Table II shows that our framework, when applied to the baseline model, enhances its cross-domain segmentation performance. The framework swaps the source and target domains and achieves consistent improvement on IOU and F1-score metrics. By aligning the conditional distribution features across domains, we achieve the same accuracy improvement for each category. Our method shows strong generalization for different cross-domain settings.

2) Changing From Geographic Locations and Imaging Modes: The PRGB_to_V and V_to_PRGB tasks involve two datasets from different imaging sensors. They have domain differences in location, resolution, illumination, and imaging patterns. These two tasks have larger domain differences and more adaptation challenges than the PIRRG_to_V task. We report the experimental results for both tasks in Tables III and IV. Our proposed JDAF surpasses all existing UDA semantic segmentation methods in the HRS images. This demonstrates the generalization and effectiveness of our framework and the importance of feature alignment of joint distributions.

In addition, JDAF performs worse than DualGAN [32] in predicting the tree category on the V_to_PRGB task. This may result from the source domain model overfitting the red bush in the IR-R-G channel of the Vaihingen data and failing to learn general features. The tree categories in the two datasets are

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TABLE II
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT UDA MODELS ON THE V_TO_PIRRG TASK (%)

Method	Clutter/ background		Impervious surfaces		Car		Tree		Low vegetation		Building		Overall	
	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	mIOU	F1-Score
Baseline (Deeplab-V3+) [13] (2018)	9.30	16.86	49.18	65.93	38.51	55.60	7.67	14.24	29.32	45.34	36.96	53.97	28.49	41.99
BiSeNet only [19] (2018)	**29.01**	**44.97**	22.70	36.99	0.69	1.36	**41.56**	**58.71**	26.12	41.42	20.93	34.61	23.50	36.34
SEANet [39] (2019)	13.16	23.23	51.33	67.79	42.44	59.56	5.15	9.79	29.14	45.13	39.61	56.73	30.14	43.70
AdaptSegNet [24] (2018)	8.36	15.33	49.55	64.64	40.95	58.11	22.59	36.79	34.43	61.50	48.01	63.41	33.98	49.96
GAN-RSDA [31] (2019)	27.39	43.43	18.66	20.71	0.59	1.82	32.06	31.93	19.72	31.08	27.40	24.24	20.97	25.54
ProDA [41] (2021)	10.63	19.21	44.70	61.72	46.78	63.74	31.59	48.02	40.55	57.71	56.85	72.49	38.51	53.82
DualGAN [32] (2021)	11.48	20.56	51.01	67.53	48.49	65.31	34.98	51.82	36.50	53.48	53.37	69.59	39.30	54.71
CCDA [33] (2022)	12.31	24.59	64.39	78.59	59.35	75.08	37.55	54.60	47.17	63.27	66.44	79.84	47.87	62.66
Adaptation from Labeled Source Patches to Easy-to-adapt Target Patches														
Baseline+MDA (ours)	12.97	22.96	62.81	77.16	56.91	72.54	32.28	48.80	42.60	59.75	66.45	79.84	45.67	60.17
Adapted-Baseline+CDA (ours)	13.73	24.14	65.38	79.07	58.14	73.53	34.56	51.36	46.72	63.68	72.09	83.78	48.44	62.60
Adaptation from Labeled Source Patches and Easy-to-adapt Target Patches to Hard-to-adapt Target Patches														
Baseline+MDA+UAL (ours)	12.77	22.65	66.11	79.60	62.05	76.58	35.23	52.10	48.26	65.10	73.75	84.89	49.69	63.49
Adapted-Baseline+CDA+UAL (ours)	13.10	23.17	**67.70**	**80.74**	**63.22**	**77.47**	36.21	53.17	**51.19**	**67.72**	**76.36**	**86.59**	**51.30**	**64.81**

TABLE III
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT UDA MODELS ON THE PRGB_TO_V TASK (%)

Method	Clutter/ background		Impervious surfaces		Car		Tree		Low vegetation		Building		Overall	
	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	mIOU	F1-Score
Baseline (Deeplab-V3+) [13] (2018)	1.03	2.03	46.39	63.37	27.33	42.93	13.58	23.73	4.61	8.82	49.39	66.12	23.72	34.50
BiSeNet only [19] (2018)	0.88	1.75	36.58	53.57	4.81	9.17	7.74	14.37	18.73	31.55	17.30	29.49	14.34	23.30
SEANet [39] (2019)	7.05	12.92	41.84	59.00	31.72	48.16	21.86	35.82	20.83	34.48	56.80	72.44	30.02	43.80
AdaptSegNet [24] (2018)	2.99	5.81	51.26	67.77	10.25	18.54	51.51	68.02	12.75	22.61	60.72	75.55	31.58	43.05
GAN-RSDA [31] (2019)	2.27	4.43	34.55	51.35	9.90	18.00	24.94	39.89	22.58	36.83	40.64	57.79	22.48	34.70
ProDA [41] (2021)	2.39	5.09	49.04	66.11	31.56	48.16	49.11	65.86	32.44	49.06	68.94	81.89	38.91	52.70
DualGAN [32] (2021)	3.94	13.88	46.19	61.33	40.31	57.88	**55.82**	**70.66**	27.85	42.17	65.44	83.00	39.93	54.82
CCDA [33] (2022)	12.38	21.55	**64.47**	77.76	43.43	60.05	52.83	69.62	38.37	55.94	**76.87**	**86.95**	48.06	61.98
Adaptation from Labeled Source Patches to Easy-to-adapt Target Patches														
Baseline+MDA (ours)	10.87	19.61	62.34	76.80	41.18	58.33	48.26	65.10	38.79	55.90	75.19	85.84	46.10	60.26
Adapted-Baseline+CDA (ours)	22.37	36.56	62.61	77.01	45.22	**65.28**	53.02	69.29	40.95	58.11	74.14	85.15	49.72	64.73
Adaptation from Labeled Source Patches and Easy-to-adapt Target Patches to Hard-to-adapt Target Patches														
Baseline+MDA+UAL (ours)	30.93	47.25	63.70	77.83	**46.25**	63.25	53.17	69.42	41.42	58.58	75.28	85.90	51.79	67.04
Adapted-Baseline+CDA+UAL (ours)	**32.71**	**49.30**	64.33	**78.29**	45.87	62.90	51.99	68.41	**42.16**	**59.31**	75.53	86.06	**52.10**	**67.38**

very different (the Postdam dataset has green-brown bushes); thus, style-transfer methods like DualGAN [32], which adapt at the image level, can handle such problems better.

D. Ablation Experiments

To analyze the validity of each design choice, we conducted ablation studies in two aspects: structural ablation and parameter ablation.

1) Structural Ablation: We tested each component on the PIRRG_to_V adaptation task of semantic segmentation and reported the results in Table V. JDAF consists of the MDA and CDA modules and the UAL method to improve easy-to-hard adaptation. MDA aligns the marginal distribution features between two domains. CDA has two components: CFA and DCA. The discriminator D_M of CFA aligns the conditional distribution features of two domains by adversarial training of the category-invariant features. DCA aggregates the dataset-level context to the features for better category prediction. The MDA + CFA means inserting the CFA block into the MDA module to achieve the extraction of category-invariant features and the alignment of conditional distributions between two domains, and the MDA + DCA means inserting the extraction of category-invariant features and DCA block into the MDA module without the D_M. The MDA + CDA means implementing the joint distribution alignment framework without the UAL method, and the MDA + CDA + UAL denotes inserting the proposed UAL method into the joint distribution alignment framework, which is our complete JDAF. The experiments are performed on the PIRRG_to_V task.

Table V shows that the MDA module improves the mIOU by 20.39% compared to the source-only model by adapting the marginal distributions between the two domains. This confirms the effectiveness of MDA in UDA semantic segmentation. Adding the CFA block (containing the D_M discriminator) to the MDA module improves the prediction accuracy for all categories except Tree. Besides, we embed the DCA block into the framework to make better use of global contextual information. The mIOU improves by 1.09% compared to the MDA-only model. Applying the joint distribution alignment framework MDA + CDA further improves the mIOU compared to MDA + CFA and MDA + DCA tasks. This shows that the joint distribution alignment helps DCA aggregate features more accurately. Applying our UAL method to the joint distribution alignment framework, which is the full JDAF, improves the segmentation accuracy for all categories except

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TABLE IV
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT UDA MODELS ON THE V_TO_PRGB TASK (%)

Method	Clutter/ background		Impervious surfaces		Car		Tree		Low vegetation		Building		Overall	
	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	mIOU	F1-Score
Baseline (Deeplab-V3+) [13] (2018)	6.99	13.04	42.98	60.12	38.01	55.08	0.53	1.06	1.59	3.13	29.09	45.05	19.86	29.58
BiSeNet only [19] (2018)	23.66	38.26	17.74	30.12	0.99	1.95	32.67	49.24	18.42	31.11	12.64	22.43	17.69	28.85
SEANet [39] (2019)	4.68	8.75	28.90	44.77	44.77	61.52	5.17	9.82	8.70	15.88	36.24	53.20	21.36	32.32
AdaptSegNet [24] (2018)	6.11	11.50	37.66	59.55	42.31	55.95	30.71	45.41	15.10	25.81	54.25	70.31	31.02	44.75
GAN-RSDA [31] (2019)	**27.75**	**43.43**	11.56	20.71	0.92	1.82	19.00	31.93	18.40	31.08	13.79	24.24	15.24	25.54
ProDA [41] (2021)	11.13	20.51	44.77	62.03	41.21	59.27	30.56	46.91	35.84	52.75	46.37	63.06	34.98	50.76
DualGAN [32] (2021)	13.56	23.84	45.96	62.97	39.71	56.84	25.80	40.97	**41.73**	**58.87**	59.01	74.22	37.63	52.95
CCDA [33] (2022)	13.27	23.43	57.65	73.14	56.99	72.27	35.87	52.80	29.77	45.88	65.44	79.11	43.17	57.77
Adaptation from Labeled Source Patches to Easy-to-adapt Target Patches														
Baseline+MDA (ours)	14.22	24.89	56.10	71.87	51.02	67.57	36.19	53.15	18.32	30.96	64.59	78.49	40.07	54.49
Adapted-Baseline+CDA (ours)	17.53	29.83	58.49	73.81	56.72	72.39	38.92	56.04	22.64	36.92	67.27	80.44	43.60	58.24
Adaptation from Labeled Source Patches and Easy-to-adapt Target Patches to Hard-to-adapt Target Patches														
Baseline+MDA+UAL (ours)	18.73	31.56	59.53	74.63	57.62	73.11	**40.27**	**57.42**	23.85	38.52	70.41	82.64	45.07	45.10
Adapted-Baseline+CDA+UAL (ours)	18.09	30.63	**60.05**	**75.04**	**58.64**	**73.93**	38.74	55.84	27.79	43.49	**71.42**	**83.33**	**45.79**	**60.38**

TABLE V
QUANTITATIVE EVALUATION RESULTS OF STRUCTURAL ABLATION ON THE PIRRG_TO_V TASK (%), WHERE BL DENOTES BASELINE

Method					Clutter/ background		Impervious surfaces		Car		Tree		Low vegetation		Building		Overall	
BL	MDA	CDA		UAL														
		CFA	DCA		IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	IOU	F1-Score	mIOU	F1-Score
✓					5.71	10.79	35.84	52.73	20.27	33.70	54.95	70.92	17.88	30.26	51.59	68.06	31.04	44.40
✓	✓				24.26	39.05	66.07	79.57	42.06	59.22	57.57	73.07	42.95	60.09	75.64	86.13	51.43	66.19
✓	✓	✓			26.16	41.47	66.17	79.64	42.66	59.80	54.23	70.33	46.89	63.85	76.09	86.42	52.03	66.92
✓	✓		✓		26.59	42.01	66.14	79.62	42.57	59.71	57.30	72.85	46.67	63.64	75.85	86.27	52.52	67.35
✓	✓	✓	✓		33.98	50.73	67.27	80.43	43.61	60.73	54.02	70.15	46.64	63.61	76.74	86.84	53.71	68.75
✓		✓	✓	✓	32.88	49.49	66.01	79.53	43.37	60.50	55.85	71.67	44.92	61.99	75.64	86.13	53.21	68.29
✓	✓			✓	23.61	38.24	68.74	81.43	42.31	59.29	58.01	73.66	44.27	61.29	76.98	86.35	52.43	66.89
✓	✓	✓		✓	31.20	47.57	67.68	80.73	**44.42**	**61.52**	54.47	70.53	47.19	64.12	75.59	86.10	53.43	68.43
✓	✓		✓	✓	29.16	45.16	67.02	80.26	42.55	59.69	56.00	71.79	45.76	62.79	76.46	86.66	52.95	67.85
✓	✓	✓	✓	✓	**38.65**	**55.75**	**68.76**	**81.49**	42.76	59.90	**58.38**	**73.72**	**47.39**	**64.30**	**77.19**	**87.13**	**55.52**	**70.38**

TABLE VI
PARAMETRIC ANALYSIS OF HIGH-CONFIDENCE THRESHOLDS σ, HYPERPARAMETERS λ_1 AND λ_2 IN PIXELS ON THE PIRRG_TO_V TASK

σ	0	0.1	0.3	0.5	0.8
mIOU	53.98	53.56	**55.52**	54.73	54.49
F1-Score	68.70	68.66	**70.38**	69.65	69.40
λ_1	1	3	4	7	9
mIOU	53.11	53.69	**55.52**	54.18	53.77
F1-Score	68.22	68.66	**70.38**	68.95	68.41
λ_2	0.5	1	2	5	7
mIOU	54.19	54.97	**55.52**	53.35	53.52
F1-Score	69.31	69.66	**70.38**	68.15	68.20

Car compared to the MDA + CDA method. The mIOU improves by 1.81%, confirming that the proposed UAL method helps the model learn the correct category prediction features. Compared to the other components (MDA and UAL), the accuracy of the model drops the most when the whole JDAF framework lacks CDA (i.e., no CDA). Besides, compared with the JDAF without CDA (here, more precisely, the JDAF without CFA), the complete JDAF has significantly improved accuracy in all categories, and mIOU is improved by 2.57%, which indicates that the addition of CDA can effectively align the features of different categories between the source and target domains.

2) Parameter Ablation: First, the semantic segmentation network selects pixels in the pseudo label $\hat{y}_t$ with higher confidence than the threshold σ to update the category-invariant features in the target domain. Second, the hyperparameters λ_1 and λ_2 are used in (22) and (23), respectively, and their purpose and effect is to further control the contribution of the loss of the high-confidence samples and the low-confidence samples to the overall loss function. The experimental results are shown in Table VI. It can be concluded that our method achieves the best mIoU when σ, λ_1, and λ_2 are set to 0.3, 5, and 2, respectively. Our method used this parameter setting on all the datasets and achieved good experimental results, which also shows that our method is not sensitive to the setting of these hyperparameters within a certain range.

E. Discussion

1) Generalizability Study: We tested the proposed JDAF on the LoveDA dataset [45] and the CITY-OSM dataset [46] to further validate its high generalizability. These datasets have large interdomain distribution differences due to the differences in architectural styles, light conditions, climates, etc. The LoveDA dataset consists of seven categories, namely background, building, road, water, barren, forest, and agriculture, which are obtained from both urban and rural areas. The CITY-OSM dataset includes only three categories: background, road, and building, which are extracted from two different cities, Chicago and Paris. As shown in Table VII, our JDAF achieves excellent results on the LoveDA dataset, outperforming other methods by 2% on both tasks. Table VIII

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TABLE VII
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT UDA MODELS ON LOVEDA TASK (%)

Task	Method	IoU(%) Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU(%)
Rural → Urban	AdaptSegNet [24] (2018)	42.35	23.73	15.61	**81.95**	13.62	28.70	22.05	32.68
	FADA [21] (2019)	43.89	12.62	12.76	80.37	12.70	32.76	24.79	31.41
	CLAN [25] (2019)	43.41	25.42	13.75	79.25	13.71	30.44	25.8	33.11
	PyCDA [40] (2019)	38.04	35.86	**45.51**	74.87	7.71	**40.39**	11.39	36.25
	CCDA [33] (2022)	46.66	42.09	37.68	61.82	33.45	36.65	44.79	43.31
	JDAF (ours)	**49.75**	**42.62**	39.55	70.79	**36.12**	39.71	**51.29**	**47.12**
Urban → Rural	AdaptSegNet [24] (2018)	26.89	40.53	30.65	50.09	16.97	32.51	28.25	32.27
	FADA [21] (2019)	24.39	32.97	25.61	47.59	15.34	34.35	20.29	28.65
	CLAN [25] (2019)	22.93	**44.78**	25.99	46.81	10.54	37.21	24.45	30.39
	PyCDA [40] (2019)	12.36	38.11	20.45	**57.16**	18.32	36.71	41.90	32.14
	CCDA [33] (2022)	47.70	43.33	39.58	51.98	21.44	57.08	56.16	45.32
	JDAF (ours)	**48.82**	43.11	**45.24**	50.40	**27.33**	**58.02**	**58.23**	**47.31**

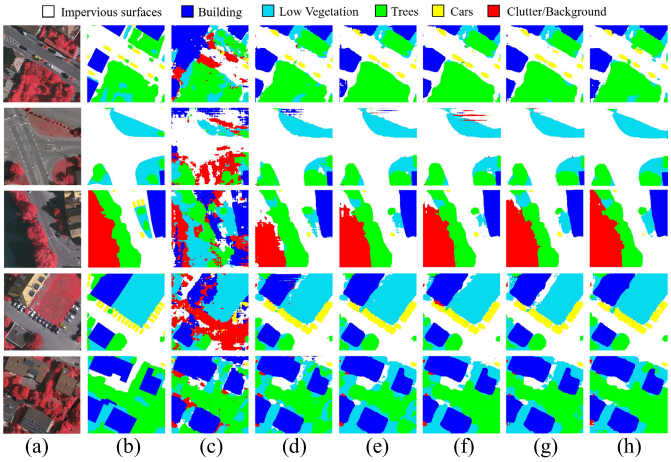


Fig. 4. Visual segmentation results for patches of the Vaihingen dataset in the PIRRG_to_V task under different ablation structures. (a) Vaihingen IR-R-G images. (b) Ground Truth. (c) Source-only. (d) MDA. (e) MDA + CFA. (f) MDA + DCA. (g) MDA + CDA. (h) JDAF.

TABLE VIII
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT UDA MODELS ON PARIS_TO_CHICAGO TASK (%)

Method	Background	Road	Building	mIOU
Baseline(Deeplab-V3) [13] (2018)	45.96	16.26	39.10	33.77
AdaptSegNet [24] (2018)	49.84	11.09	52.71	37.88
AdvEnt [26] (2019)	50.42	22.92	49.43	40.92
CCM-ERAM [44] (2022)	**54.20**	24.66	**56.32**	45.06
CCDA [33] (2022)	51.98	41.06	54.34	49.12
JDAF (ours)	51.96	**44.55**	53.80	**50.10**

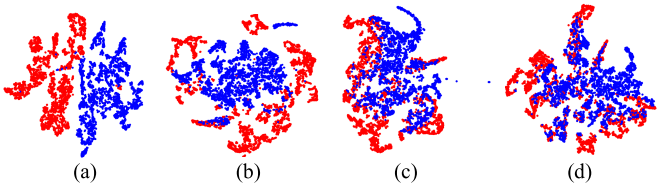


Fig. 5. Visualization results of t-SNE features under different ablation structures in the PIRRG_to_V task, where (a)–(d) represent the features extracted from Source-only, MDA module, MDA + CDA module, and JDAF. The red and blue points denote the features from the source and the target domains, respectively.

shows that our JDAF also achieves the SOTA results on the CITY-OSM dataset.

Compared to JDAF performance on the Postdam (six categories), Vaihingen (six categories) tasks, and LoveDA (seven categories) tasks, on the CITY-OSM (only three categories) task, accuracy was not remarkably improved compared to other methods (especially CCDA [33]). Since the CITY-OSM dataset only contains three types of land surfaces, few categories make it easier to achieve the feature alignment of different categories by only achieving MDA, so the benefits of introducing CDA are not obvious; however, for datasets with a relatively large number of categories (such as the Potsdam, Vaihingen, and LoveDA datasets), by only achieving MDA, it is more difficult to achieve the feature alignment of different categories, and therefore, it is necessary to realize the alignment of conditional distributions at the same time, that is, the alignment of joint distributions needs to be realized. In particular, the greater the number of categories in a dataset, the greater the benefits of CDA.

2) Qualitative Comparison Study: The quantitative adaptation results on the PIRRG_to_V task can be viewed in Fig. 4. We selected five representative images to cover different categories. The proposed JDAF has achieved good results in some confusing categories. The second row describes the improvement of low vegetation. From the third row to the fifth row, it can be seen that the segmentation of trees and low vegetation is significantly improved, although there are some overlaps between them. Meanwhile, the first row shows that the tiny trees between the two cars are segmented accurately, owing to the alignment of conditional distributions.

3) Visualization of Feature Distributions: We use t-SNE to visualize the features of four models, deeplab-v3+, MDA, MDA + CDA, and JDAF, to examine the effects of different structures on the category decision boundaries. The visualization results are demonstrated in Fig. 5. Without any adaptation of the source domain model, the feature distributions of the source and target domains are highly different, and the prediction of model is almost random, which shows that the data distribution differences between the source and target domains are extremely large. These tasks are difficult to adapt, which is also the reason why Postdam and Vaihingen become common benchmarks for UDA tasks of remote sensing datasets. Besides, Fig. 5(b) shows that aligning only the interdomain marginal distributions just narrows the distributions between the features of the two data sets. As can be observed from Fig. 5(c) and (d), the proposed JDAF performs interdomain

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joint distribution alignment of features, adaptively aligning the conditional distributions based on the alignment of marginal distributions so that the category decision boundaries of the target domain are best aligned with those of the source domain, which helps reduce the risk of negative transfer and ensures tighter feature matching results in the source and target domains.

V. Conclusion

In this article, we have proposed an adaptive framework for the alignment of joint distributions between domains, namely JDAF, to transfer label-rich knowledge from the source domain to the unlabeled target domain. The network structure of JDAF mainly consists of two modules: the MDA module and the CDA module. The MDA module is used to align the interdomain marginal distributions from local to global. For the CDA module, the CFA block is used to align the conditional distributions by adversarially training category-invariant features that are shared between the two domains, and the DCA block is designed to augment the original pixel representation and improve intraclass compactness and interclass dispersion at the dataset level. Besides, the UAL method, consisting of uncertainty-adaptive self-supervised learning and adversarial learning, is proposed to further fit the distribution of the target domain for hard-to-adapt samples. The proposed JDAF can improve segmentation accuracy for all categories without sacrificing accuracy for a single category. Extensive experimental results and insightful analyses have demonstrated that our approach outperforms the existing state-of-the-art methods.

To extract the category-invariant features of the target domain, the source domain model needs to adapt to the target domain first, which requires MDA before CDA. Our future work may aim to align both distributions simultaneously to achieve better adaptation and a faster training process.

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